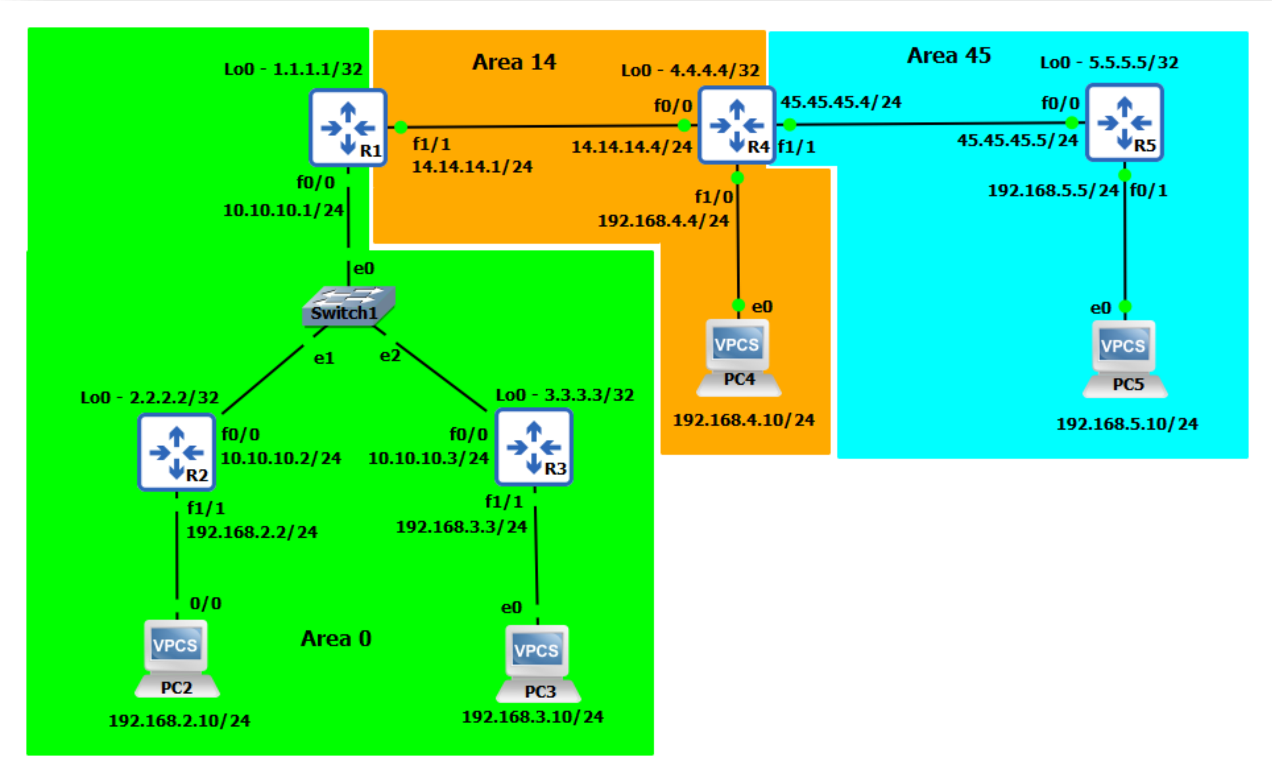
OSPFv2 – Multi-Area



Task

1. Configure routers with IP address as shown in topology and configure enable password as **ccna** & device access password as **cisco (Refer Lab 11)**
2. Configure Multi-Area OSPF process as per the topology.
3. Configure Virtual-link to provide communication of Area 45 with Area 0
4. Configure Priority of 255 to make R1 as permanent DR.

Task -2: Configure Multi-Area OSPF process as per the topology.

R1 OSPF Configuration:

R1#config t
R1(config)#router ospf 1
R1(config-router)#network 14.14.14.1 0.0.0.0 area 14
R1(config-router)#network 10.10.10.0 0.0.0.255 area 0
R1(config-router)#network 1.1.1.1 0.0.0.0 area 0
R1(config-router)#exit
R1(config)#exit
R1#

R2 OSPF Configuration:

R2#config t
R2(config)#router ospf 2

R2(config-router)#network 192.168.2.0 0.0.0.255 area 0
R2(config-router)#network 10.10.10.0 0.0.0.255 area 0
R2(config-router)#network 2.2.2.2 0.0.0.0 area 0
R2(config-router)#exit
R2(config)#exit
R2#

R3 OSPF Configuration:

R3#config t
R3(config)#router ospf 3
R3(config-router)#network 192.168.3.0 0.0.0.255 area 0
R3(config-router)#network 10.10.10.0 0.0.0.255 area 0
R3(config-router)#network 3.3.3.3 0.0.0.0 area 0
R3(config-router)#exit
R3(config)#exit
R3#

R4 OSPF Configuration:

R4#config t
R4(config)#router ospf 4
R4(config-router)#network 192.168.4.0 0.0.0.255 area 14
R4(config-router)#network 14.14.14.4 0.0.0.0 area 14
R4(config-router)#network 45.45.45.4 0.0.0.0 area 45
R4(config-router)#network 4.4.4.4 0.0.0.0 area 14
R4(config-router)#exit
R4(config)#exit
R4#

R5 OSPF Configuration:

R5#config t
R5(config)#router ospf 5
R5(config-router)#network 192.168.5.0 0.0.0.255 area 45
R5(config-router)#network 45.45.45.5 0.0.0.0 area 45
R5(config-router)#network 5.5.5.5 0.0.0.0 area 45
R5(config-router)#exit
R5(config)#exit
R5#

✓ Verification & Testing:

R1#sh ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  It is an area border router

```
  Number of areas in this router is 2. 2 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    1.1.1.1 0.0.0.0 area 0
    10.10.10.0 0.0.0.255 area 0
    14.14.14.1 0.0.0.0 area 14
  Routing Information Sources:
    Gateway         Distance      Last Update
    2.2.2.2              110          00:00:22
    4.4.4.4              110          00:00:22
    3.3.3.3              110          00:00:32
  Distance: (default is 110)
```

R2#sh ip protocols

```
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 2"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    2.2.2.2 0.0.0.0 area 0
    10.10.10.0 0.0.0.255 area 0
    192.168.2.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110          00:00:31
    3.3.3.3              110          00:00:41
  Distance: (default is 110)
```

R3#sh ip protocols

```
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 3"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.3
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    3.3.3.3 0.0.0.0 area 0
    10.10.10.0 0.0.0.255 area 0
    192.168.3.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110          00:00:33
    2.2.2.2              110          00:00:44
  Distance: (default is 110)
```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R4#sh ip protocols

```
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 4"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 4.4.4.4
  Number of areas in this router is 2. 2 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    4.4.4.4 0.0.0.0 area 14
    14.14.14.4 0.0.0.0 area 14
    45.45.45.4 0.0.0.0 area 45
    192.168.4.0 0.0.0.255 area 14
  Routing Information Sources:
    Gateway         Distance      Last Update
    5.5.5.5              110          00:00:18
    1.1.1.1              110          00:00:36
  Distance: (default is 110)
```

R5#sh ip protocols

```
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 5"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 5.5.5.5
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    5.5.5.5 0.0.0.0 area 45
    45.45.45.5 0.0.0.0 area 45
    192.168.5.0 0.0.0.255 area 45
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 110)
```

R1#sh ip route ospf

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/2] via 10.10.10.2, 00:02:10, FastEthernet0/0
```

```
     3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/2] via 10.10.10.3, 00:02:20, FastEthernet0/0
     4.0.0.0/32 is subnetted, 1 subnets
O       4.4.4.4 [110/2] via 14.14.14.4, 00:02:10, FastEthernet1/1
O    192.168.2.0/24 [110/2] via 10.10.10.2, 00:02:10, FastEthernet0/0
O    192.168.3.0/24 [110/2] via 10.10.10.3, 00:02:20, FastEthernet0/0
O    192.168.4.0/24 [110/2] via 14.14.14.4, 00:02:10, FastEthernet1/1
```

R2#sh ip route ospf

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

     1.0.0.0/32 is subnetted, 1 subnets
O       1.1.1.1 [110/2] via 10.10.10.1, 00:03:40, FastEthernet0/0
     3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/2] via 10.10.10.3, 00:03:50, FastEthernet0/0
     4.0.0.0/32 is subnetted, 1 subnets
O IA    4.4.4.4 [110/3] via 10.10.10.1, 00:03:40, FastEthernet0/0
     14.0.0.0/24 is subnetted, 1 subnets
O IA    14.14.14.0 [110/2] via 10.10.10.1, 00:03:40, FastEthernet0/0
O    192.168.3.0/24 [110/2] via 10.10.10.3, 00:03:50, FastEthernet0/0
O IA 192.168.4.0/24 [110/3] via 10.10.10.1, 00:03:40, FastEthernet0/0
```

R3#sh ip route ospf

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

     1.0.0.0/32 is subnetted, 1 subnets
O       1.1.1.1 [110/2] via 10.10.10.1, 00:03:53, FastEthernet0/0
     2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/2] via 10.10.10.2, 00:03:53, FastEthernet0/0
     4.0.0.0/32 is subnetted, 1 subnets
O IA    4.4.4.4 [110/3] via 10.10.10.1, 00:03:43, FastEthernet0/0
     14.0.0.0/24 is subnetted, 1 subnets
O IA    14.14.14.0 [110/2] via 10.10.10.1, 00:03:53, FastEthernet0/0
```

```
O     192.168.2.0/24 [110/2] via 10.10.10.2, 00:03:53, FastEthernet0/0
O IA  192.168.4.0/24 [110/3] via 10.10.10.1, 00:03:43, FastEthernet0/0
```

R4#sh ip route ospf

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O IA     1.1.1.1 [110/2] via 14.14.14.1, 00:03:45, FastEthernet0/0
      2.0.0.0/32 is subnetted, 1 subnets
O IA     2.2.2.2 [110/3] via 14.14.14.1, 00:03:44, FastEthernet0/0
      3.0.0.0/32 is subnetted, 1 subnets
O IA     3.3.3.3 [110/3] via 14.14.14.1, 00:03:45, FastEthernet0/0
      5.0.0.0/32 is subnetted, 1 subnets
O        5.5.5.5 [110/2] via 45.45.45.5, 00:03:27, FastEthernet1/1
      10.0.0.0/24 is subnetted, 1 subnets
O IA     10.10.10.0 [110/2] via 14.14.14.1, 00:03:45, FastEthernet0/0
O IA  192.168.2.0/24 [110/3] via 14.14.14.1, 00:03:44, FastEthernet0/0
O IA  192.168.3.0/24 [110/3] via 14.14.14.1, 00:03:45, FastEthernet0/0
O     192.168.5.0/24 [110/2] via 45.45.45.5, 00:03:27, FastEthernet1/1
```

R5#sh ip route ospf

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

  ----   No OSPF Routes -----
R5#
```

R1#sh ip ospf neighbor

```
Neighbor ID   Pri   State        Dead Time   Address       Interface
2.2.2.2         1   FULL/BDR     00:00:35    10.10.10.2    FastEthernet0/0
3.3.3.3         1   FULL/DR      00:00:34    10.10.10.3    FastEthernet0/0
4.4.4.4         1   FULL/DR      00:00:32    14.14.14.4    FastEthernet1/1
R1#
```

R2#sh ip ospf neighbor

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DROTHER    00:00:38    10.10.10.1      FastEthernet0/0
3.3.3.3           1   FULL/DR         00:00:30    10.10.10.3      FastEthernet0/0
R2#
```

R3#sh ip ospf neighbor

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DROTHER    00:00:36    10.10.10.1      FastEthernet0/0
2.2.2.2           1   FULL/BDR        00:00:39    10.10.10.2      FastEthernet0/0
R3#
```

R4#sh ip ospf neighbor

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/BDR        00:00:37    14.14.14.1      FastEthernet0/0
5.5.5.5           1   FULL/DR         00:00:32    45.45.45.5      FastEthernet1/1
R4#
```

R5#sh ip ospf neighbor

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
4.4.4.4           1   FULL/BDR        00:00:35    45.45.45.4      FastEthernet0/0
R5#
```

R1#sh ip ospf database

```
            OSPF Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         819         0x80000002 0x00CD06 2
2.2.2.2         2.2.2.2         820         0x80000002 0x002724 3
3.3.3.3         3.3.3.3         820         0x80000002 0x003805 3

                Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
10.10.10.3      3.3.3.3         820         0x80000001 0x00E20E

                Summary Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
4.4.4.4         1.1.1.1         805         0x80000001 0x00C660
14.14.14.0      1.1.1.1         870         0x80000001 0x007B92
192.168.4.0     1.1.1.1         805         0x80000001 0x00A128

                Router Link States (Area 14)
```

```
Link ID         ADV Router      Age         Seq#        Checksum Link count
1.1.1.1         1.1.1.1         810         0x80000002 0x0022AD 1
4.4.4.4         4.4.4.4         811         0x80000002 0x0071A4 3

                Net Link States (Area 14)

Link ID         ADV Router      Age         Seq#        Checksum
14.14.14.4      4.4.4.4         811         0x80000001 0x00BF28

                Summary Net Link States (Area 14)

Link ID         ADV Router  Age         Seq#        Checksum
1.1.1.1         1.1.1.1     870         0x80000001 0x0047EC
2.2.2.2         1.1.1.1     805         0x80000001 0x00230C
3.3.3.3         1.1.1.1     815         0x80000001 0x00F436
10.10.10.0      1.1.1.1     870         0x80000001 0x000C0E
192.168.2.0     1.1.1.1     805         0x80000001 0x00B714
192.168.3.0     1.1.1.1     815         0x80000001 0x00AC1E
R1#
```

R2#sh ip ospf database

```
            OSPF Router with ID (2.2.2.2) (Process ID 2)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#        Checksum Link count
1.1.1.1         1.1.1.1         855         0x80000002 0x00CD06      2
2.2.2.2         2.2.2.2         854         0x80000002 0x002724      3
3.3.3.3         3.3.3.3         855         0x80000002 0x003805      3

                Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#        Checksum
10.10.10.3      3.3.3.3         855         0x80000001 0x00E20E

                Summary Net Link States (Area 0)

Link ID         ADV Router  Age         Seq#            Checksum
4.4.4.4         1.1.1.1     840         0x80000001      0x00C660
14.14.14.0      1.1.1.1     906         0x80000001      0x007B92
192.168.4.0     1.1.1.1     840         0x80000001      0x00A128
R2#
```

R3#sh ip ospf database

```
            OSPF Router with ID (3.3.3.3) (Process ID 3)

                Router Link States (Area 0)
```

```
Link ID         ADV Router      Age         Seq#           Checksum Link count
1.1.1.1          1.1.1.1        857       0x80000002 0x00CD06      2
2.2.2.2          2.2.2.2        857       0x80000002 0x002724      3
3.3.3.3          3.3.3.3        857       0x80000002 0x003805      3

                Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#           Checksum
10.10.10.3      3.3.3.3         857       0x80000001 0x00E20E

                Summary Net Link States (Area 0)

Link ID         ADV Router      Age       Seq#       Checksum
4.4.4.4          1.1.1.1        843     0x80000001 0x00C660
14.14.14.0       1.1.1.1        908     0x80000001 0x007B92
192.168.4.0      1.1.1.1        843     0x80000001 0x00A128
R3#
```

R4#sh ip ospf database

```
            OSPF Router with ID (4.4.4.4) (Process ID 4)

                Router Link States (Area 14)

Link ID         ADV Router      Age         Seq#           Checksum Link count
1.1.1.1          1.1.1.1        851       0x80000002 0x0022AD 1
4.4.4.4          4.4.4.4        850       0x80000002 0x0071A4 3

                Net Link States (Area 14)

Link ID         ADV Router      Age         Seq#           Checksum
14.14.14.4      4.4.4.4         850       0x80000001 0x00BF28

                Summary Net Link States (Area 14)

Link ID         ADV Router    Age         Seq#       Checksum
1.1.1.1          1.1.1.1        910     0x80000001 0x0047EC
2.2.2.2          1.1.1.1        845     0x80000001 0x00230C
3.3.3.3          1.1.1.1        855     0x80000001 0x00F436
10.10.10.0       1.1.1.1        910     0x80000001 0x000C0E
192.168.2.0      1.1.1.1        845     0x80000001 0x00B714
192.168.3.0      1.1.1.1        855     0x80000001 0x00AC1E

                Router Link States (Area 45)

Link ID         ADV Router      Age         Seq#           Checksum Link count
4.4.4.4          4.4.4.4        831       0x80000002 0x00B940 1
5.5.5.5          5.5.5.5        832       0x80000002 0x00A4A7 3

                Net Link States (Area 45)
```

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```
Link ID        ADV Router     Age        Seq#       Checksum
45.45.45.5     5.5.5.5        832        0x80000001 0x00EF85
R4#
```

R5#sh ip ospf database

```
            OSPF Router with ID (5.5.5.5) (Process ID 5)

                Router Link States (Area 45)

Link ID        ADV Router  Age        Seq#       Checksum Link count
4.4.4.4        4.4.4.4     837        0x80000002 0x00B940 1
5.5.5.5        5.5.5.5     837        0x80000002 0x00A4A7 3

                Net Link States (Area 45)

Link ID        ADV Router     Age        Seq#       Checksum
45.45.45.5     5.5.5.5        837        0x80000001 0x00EF85
R5#
```

Ping Loopback IP of all other routers from R1

R1#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/16/20 ms

R1#ping 3.3.3.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/16/24 ms

R1#ping 4.4.4.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/19/24 ms

R1#ping 5.5.5.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1#

Ping Loopback IP of all other routers from R2

R2#ping 1.1.1.1
Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/16/20 ms

R2#ping 3.3.3.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/17/24 ms

R2#ping 4.4.4.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/43/44 ms

R2#ping 5.5.5.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R2#

Ping Loopback IP of all other routers from R3

R3#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/17/24 ms

R3#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/16/20 ms

R3#ping 4.4.4.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/40/44 ms

R3#ping 5.5.5.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R3#

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Ping Loopback IP of all other routers from R4

R4#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/18/20 ms

R4#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 36/40/44 ms

R4#ping 3.3.3.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/37/44 ms

R4#ping 5.5.5.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/17/28 ms

Ping Loopback IP of all other routers from R5

R5#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R5#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R5#ping 3.3.3.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R5#ping 4.4.4.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Task 3: Configure Virtual-link to provide communication between Area 45 and Area 0.

R1 Router Configuration:

```
R1#config t
R1(config)#router ospf 1
R1(config-router)#area 14 virtual-link 4.4.4.4
R1(config-router)#exit
R1(config)#exit
R1#
```

R2 Router Configuration:

```
R4#config t
R4(config)#router ospf 4
R4(config-router)#area 14 virtual-link 1.1.1.1
R4(config-router)##exit
R4(config)#exit
R4#
*Oct 17 16:16:38.203: %OSPF-5-ADJCHG: Process 4, Nbr 1.1.1.1 on OSPF_VL0 from LOADING to FULL, Loading Done
```

- ✓ **Verification & Testing:**

R1#sh ip ospf virtual-links

```
Virtual Link OSPF_VL0 to router 4.4.4.4 is up
  Run as demand circuit
  DoNotAge LSA allowed.
  Transit area 14, via interface FastEthernet1/1
 Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Transmit Delay is 1 sec, State POINT_TO_POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:02
    Adjacency State FULL (Hello suppressed)
    Index 3/4, retransmission queue length 0, number of retransmission 0
    First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
    Last retransmission scan length is 0, maximum is 0
    Last retransmission scan time is 0 msec, maximum is 0 msec
R1#
```

R4#sh ip ospf virtual-links

```
Virtual Link OSPF_VL0 to router 1.1.1.1 is up
  Run as demand circuit
  DoNotAge LSA allowed.
  Transit area 14, via interface FastEthernet0/0
 Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           1         no          no            Base
  Transmit Delay is 1 sec, State POINT_TO_POINT,
```

```
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:03
    Adjacency State FULL (Hello suppressed)
    Index 1/3, retransmission queue length 0, number of retransmission 0
    First 0x0(0)/0x0(0) Next 0x0(0)/0x0(0)
    Last retransmission scan length is 0, maximum is 0
    Last retransmission scan time is 0 msec, maximum is 0 msec
R4#
```

R1#sh ip ospf neighbor

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
4.4.4.4           0   FULL/  -        -           14.14.14.4      OSPF_VL0
2.2.2.2           1   FULL/BDR        00:00:34    10.10.10.2      FastEthernet0/0
3.3.3.3           1   FULL/DR         00:00:38    10.10.10.3      FastEthernet0/0
4.4.4.4           1   FULL/DR         00:00:35    14.14.14.4      FastEthernet1/1
R1#
```

R4#sh ip ospf neighbor

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           0   FULL/  -        -           14.14.14.1      OSPF_VL0
1.1.1.1           1   FULL/BDR        00:00:30    14.14.14.1      FastEthernet0/0
5.5.5.5           1   FULL/DR         00:00:38    45.45.45.5      FastEthernet1/1
R4#
```

R1#sh ip route ospf | begin Gateway

```
Gateway of last resort is not set

      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/2] via 10.10.10.2, 01:30:56, FastEthernet0/0
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/2] via 10.10.10.3, 01:31:06, FastEthernet0/0
      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/2] via 14.14.14.4, 01:30:56, FastEthernet1/1
      5.0.0.0/32 is subnetted, 1 subnets
O IA     5.5.5.5 [110/3] via 14.14.14.4, 00:04:59, FastEthernet1/1
      45.0.0.0/24 is subnetted, 1 subnets
O IA     45.45.45.0 [110/2] via 14.14.14.4, 00:04:59, FastEthernet1/1
O     192.168.2.0/24 [110/2] via 10.10.10.2, 01:30:56, FastEthernet0/0
O     192.168.3.0/24 [110/2] via 10.10.10.3, 01:31:06, FastEthernet0/0
O     192.168.4.0/24 [110/2] via 14.14.14.4, 01:30:56, FastEthernet1/1
O IA  192.168.5.0/24 [110/3] via 14.14.14.4, 00:04:59, FastEthernet1/1
```

R2#sh ip route ospf | begin Gateway

```
Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/2] via 10.10.10.1, 01:31:14, FastEthernet0/0
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/2] via 10.10.10.3, 01:31:24, FastEthernet0/0
```

```
      4.0.0.0/32 is subnetted, 1 subnets
O IA     4.4.4.4 [110/3] via 10.10.10.1, 00:05:07, FastEthernet0/0
      5.0.0.0/32 is subnetted, 1 subnets
O IA     5.5.5.5 [110/4] via 10.10.10.1, 00:05:07, FastEthernet0/0
      14.0.0.0/24 is subnetted, 1 subnets
O IA     14.14.14.0 [110/2] via 10.10.10.1, 01:31:14, FastEthernet0/0
      45.0.0.0/24 is subnetted, 1 subnets
O IA     45.45.45.0 [110/3] via 10.10.10.1, 00:05:07, FastEthernet0/0
O     192.168.3.0/24 [110/2] via 10.10.10.3, 01:31:24, FastEthernet0/0
O IA  192.168.4.0/24 [110/3] via 10.10.10.1, 00:05:07, FastEthernet0/0
O IA  192.168.5.0/24 [110/4] via 10.10.10.1, 00:05:07, FastEthernet0/0
```

R3#sh ip route ospf | begin Gateway

```
Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/2] via 10.10.10.1, 01:31:29, FastEthernet0/0
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/2] via 10.10.10.2, 01:31:29, FastEthernet0/0
      4.0.0.0/32 is subnetted, 1 subnets
O IA     4.4.4.4 [110/3] via 10.10.10.1, 00:05:12, FastEthernet0/0
      5.0.0.0/32 is subnetted, 1 subnets
O IA     5.5.5.5 [110/4] via 10.10.10.1, 00:05:12, FastEthernet0/0
      14.0.0.0/24 is subnetted, 1 subnets
O IA     14.14.14.0 [110/2] via 10.10.10.1, 01:31:29, FastEthernet0/0
      45.0.0.0/24 is subnetted, 1 subnets
O IA     45.45.45.0 [110/3] via 10.10.10.1, 00:05:12, FastEthernet0/0
O     192.168.2.0/24 [110/2] via 10.10.10.2, 01:31:29, FastEthernet0/0
O IA  192.168.4.0/24 [110/3] via 10.10.10.1, 00:05:12, FastEthernet0/0
O IA  192.168.5.0/24 [110/4] via 10.10.10.1, 00:05:12, FastEthernet0/0
```

R4#sh ip route ospf | begin Gateway

```
Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/2] via 14.14.14.1, 00:05:27, FastEthernet0/0
      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/3] via 14.14.14.1, 00:05:27, FastEthernet0/0
      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/3] via 14.14.14.1, 00:05:27, FastEthernet0/0
      5.0.0.0/32 is subnetted, 1 subnets
O        5.5.5.5 [110/2] via 45.45.45.5, 00:05:37, FastEthernet1/1
      10.0.0.0/24 is subnetted, 1 subnets
O        10.10.10.0 [110/2] via 14.14.14.1, 00:05:27, FastEthernet0/0
O     192.168.2.0/24 [110/3] via 14.14.14.1, 00:05:27, FastEthernet0/0
O     192.168.3.0/24 [110/3] via 14.14.14.1, 00:05:27, FastEthernet0/0
O     192.168.5.0/24 [110/2] via 45.45.45.5, 00:05:37, FastEthernet1/1
R4#
```

R5#sh ip route ospf | begin Gateway

```
Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O IA     1.1.1.1 [110/3] via 45.45.45.4, 00:05:31, FastEthernet0/0
      2.0.0.0/32 is subnetted, 1 subnets
O IA     2.2.2.2 [110/4] via 45.45.45.4, 00:05:31, FastEthernet0/0
      3.0.0.0/32 is subnetted, 1 subnets
O IA     3.3.3.3 [110/4] via 45.45.45.4, 00:05:31, FastEthernet0/0
      4.0.0.0/32 is subnetted, 1 subnets
O IA     4.4.4.4 [110/2] via 45.45.45.4, 00:05:36, FastEthernet0/0
      10.0.0.0/24 is subnetted, 1 subnets
O IA     10.10.10.0 [110/3] via 45.45.45.4, 00:05:31, FastEthernet0/0
      14.0.0.0/24 is subnetted, 1 subnets
O IA     14.14.14.0 [110/2] via 45.45.45.4, 00:05:36, FastEthernet0/0
O IA  192.168.2.0/24 [110/4] via 45.45.45.4, 00:05:31, FastEthernet0/0
O IA  192.168.3.0/24 [110/4] via 45.45.45.4, 00:05:31, FastEthernet0/0
O IA  192.168.4.0/24 [110/2] via 45.45.45.4, 00:05:36, FastEthernet0/0
R5#
```

Ping R5's Loopback IP from R1

R1#ping 5.5.5.5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/40/64 ms
```

Ping R5's Loopback IP from R2

R2#ping 5.5.5.5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 48/57/60 ms
```

Ping R5's Loopback IP from R3

R3# ping 5.5.5.5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 48/57/64 ms
```

Ping R5's Loopback IP from R4

R4#ping 5.5.5.5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 5.5.5.5, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/16/20 ms
```

Ping Loopback IP of all other router from R5

R5#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/40/64 ms

R5#ping 2.2.2.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/56/64 ms

R5#ping 3.3.3.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 44/57/80 ms

R5#ping 4.4.4.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/19/24 ms

Ping all other PCs from PC2

PC2> ping 192.168.3.10
84 bytes from 192.168.3.10 icmp_seq=1 ttl=62 time=53.132 ms
84 bytes from 192.168.3.10 icmp_seq=2 ttl=62 time=35.169 ms
84 bytes from 192.168.3.10 icmp_seq=3 ttl=62 time=35.211 ms
84 bytes from 192.168.3.10 icmp_seq=4 ttl=62 time=35.200 ms
84 bytes from 192.168.3.10 icmp_seq=5 ttl=62 time=35.220 ms

PC2> ping 192.168.4.10
84 bytes from 192.168.4.10 icmp_seq=1 ttl=61 time=115.908 ms
84 bytes from 192.168.4.10 icmp_seq=2 ttl=61 time=58.132 ms
84 bytes from 192.168.4.10 icmp_seq=3 ttl=61 time=56.952 ms
84 bytes from 192.168.4.10 icmp_seq=4 ttl=61 time=59.035 ms
84 bytes from 192.168.4.10 icmp_seq=5 ttl=61 time=56.059 ms

PC2> ping 192.168.5.10
192.168.5.10 icmp_seq=1 timeout
192.168.5.10 icmp_seq=2 timeout
84 bytes from 192.168.5.10 icmp_seq=3 ttl=60 time=76.014 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=60 time=80.073 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=60 time=79.971 ms

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Ping all other PCs from PC3

PC3> ping 192.168.2.10
84 bytes from 192.168.2.10 icmp_seq=1 ttl=62 time=33.172 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=62 time=34.152 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=62 time=43.169 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=62 time=39.086 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=62 time=35.172 ms

PC3> ping 192.168.4.10
84 bytes from 192.168.4.10 icmp_seq=1 ttl=61 time=59.953 ms
84 bytes from 192.168.4.10 icmp_seq=2 ttl=61 time=59.071 ms
84 bytes from 192.168.4.10 icmp_seq=3 ttl=61 time=56.024 ms
84 bytes from 192.168.4.10 icmp_seq=4 ttl=61 time=55.087 ms
84 bytes from 192.168.4.10 icmp_seq=5 ttl=61 time=57.118 ms

PC3> ping 192.168.5.10
84 bytes from 192.168.5.10 icmp_seq=1 ttl=60 time=80.045 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=60 time=81.141 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=60 time=80.980 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=60 time=82.004 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=60 time=81.946 ms

Ping all other PCs from PC4

PC4> ping 192.168.2.10
84 bytes from 192.168.2.10 icmp_seq=1 ttl=61 time=57.994 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=61 time=58.975 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=61 time=59.145 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=61 time=62.122 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=61 time=66.009 ms

PC4> ping 192.168.3.10
84 bytes from 192.168.3.10 icmp_seq=1 ttl=61 time=57.174 ms
84 bytes from 192.168.3.10 icmp_seq=2 ttl=61 time=57.030 ms
84 bytes from 192.168.3.10 icmp_seq=3 ttl=61 time=59.119 ms
84 bytes from 192.168.3.10 icmp_seq=4 ttl=61 time=58.089 ms
84 bytes from 192.168.3.10 icmp_seq=5 ttl=61 time=63.982 ms

PC4> ping 192.168.5.10
84 bytes from 192.168.5.10 icmp_seq=1 ttl=62 time=38.251 ms
84 bytes from 192.168.5.10 icmp_seq=2 ttl=62 time=42.090 ms
84 bytes from 192.168.5.10 icmp_seq=3 ttl=62 time=33.132 ms
84 bytes from 192.168.5.10 icmp_seq=4 ttl=62 time=34.216 ms
84 bytes from 192.168.5.10 icmp_seq=5 ttl=62 time=33.185 ms

Ping all other PCs from PC5

PC5> ping 192.168.2.10
84 bytes from 192.168.2.10 icmp_seq=1 ttl=60 time=77.100 ms
84 bytes from 192.168.2.10 icmp_seq=2 ttl=60 time=77.034 ms

84 bytes from 192.168.2.10 icmp_seq=3 ttl=60 time=77.024 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=60 time=78.020 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=60 time=81.089 ms

PC5> ping 192.168.3.10
192.168.3.10 icmp_seq=1 timeout
192.168.3.10 icmp_seq=2 timeout
84 bytes from 192.168.3.10 icmp_seq=3 ttl=60 time=82.945 ms
84 bytes from 192.168.3.10 icmp_seq=4 ttl=60 time=87.043 ms
84 bytes from 192.168.3.10 icmp_seq=5 ttl=60 time=85.070 ms

PC5> ping 192.168.4.10
192.168.4.10 icmp_seq=1 timeout
192.168.4.10 icmp_seq=2 timeout
84 bytes from 192.168.4.10 icmp_seq=3 ttl=62 time=34.274 ms
84 bytes from 192.168.4.10 icmp_seq=4 ttl=62 time=34.168 ms
84 bytes from 192.168.4.10 icmp_seq=5 ttl=62 time=36.207 ms

Task 4: Configure Priority of 255 to make R1 as permanent DR.

R1#config t
R1(config)#int fa0/0
R1(config-if)#ip ospf priority 255
R1(config-if)#exit
R1(config)#exit
R1#

Note: Because higher priority routers will not pre-empt existing DR and BDR routers, we need to manually reset the OSPF process on DR and then on BDR router.

R3#clear ip ospf process
Reset ALL OSPF processes? [no]: Yes
R3#
***Oct 17 18:34:00.987: %OSPF-5-ADJCHG: Process 3, Nbr 1.1.1.1 on FastEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached**
***Oct 17 18:34:00.991: %OSPF-5-ADJCHG: Process 3, Nbr 2.2.2.2 on FastEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached**

***Oct 17 18:34:04.927: %OSPF-5-ADJCHG: Process 3, Nbr 1.1.1.1 on FastEthernet0/0 from LOADING to FULL, Loading Done**
***Oct 17 18:34:04.927: %OSPF-5-ADJCHG: Process 3, Nbr 2.2.2.2 on FastEthernet0/0 from LOADING to FULL, Loading Done**

R2#clear ip ospf process
Reset ALL OSPF processes? [no]: Yes
R2#
***Oct 17 18:34:20.299: %OSPF-5-ADJCHG: Process 2, Nbr 1.1.1.1 on FastEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached**
***Oct 17 18:34:20.303: %OSPF-5-ADJCHG: Process 2, Nbr 3.3.3.3 on FastEthernet0/0 from FULL to DOWN, Neighbor Down: Interface down or detached**

***Oct 17 18:34:29.075: %OSPF-5-ADJCHG: Process 2, Nbr 1.1.1.1 on FastEthernet0/0 from LOADING to FULL, Loading Done**
***Oct 17 18:34:33.371: %OSPF-5-ADJCHG: Process 2, Nbr 3.3.3.3 on FastEthernet0/0 from LOADING to FULL, Loading Done**

- ✓ **Verification & Testing:**

R1#sh ip ospf neighbor

```
Neighbor ID     Pri   State              Dead Time   Address        Interface
4.4.4.4           0   FULL/  -        -              14.14.14.4     OSPF_VL0
2.2.2.2           1   FULL/DROTHER       00:00:36    10.10.10.2     FastEthernet0/0
3.3.3.3           1   FULL/BDR           00:00:39    10.10.10.3     FastEthernet0/0
4.4.4.4           1   FULL/DR            00:00:38    14.14.14.4     FastEthernet1/1
R1#
```

R2#sh ip ospf neighbor

```
Neighbor ID     Pri        State        Dead Time   Address        Interface
1.1.1.1         255   FULL/DR           00:00:33    10.10.10.1     FastEthernet0/0
3.3.3.3           1   FULL/BDR          00:00:38    10.10.10.3     FastEthernet0/0
R2#
```

R3#sh ip ospf neighbor

```
Neighbor ID     Pri   State              Dead Time   Address        Interface
1.1.1.1         255   FULL/DR            00:00:34    10.10.10.1     FastEthernet0/0
2.2.2.2           1   FULL/DROTHER       00:00:32    10.10.10.2     FastEthernet0/0
R3#
```

Important Commands:

sh ip int brief | exclude unassign
sh ip protocols
sh ip ospf interface brief
show ip route ospf
sh ip ospf neighbor
sh ip ospf database
sh ip ospf interface xx
sh ip ospf virtual-links
clear ip ospf process

info@rsatechforum.in